Applications of quadratic functions

1

a $t = 2, t = 3$ **b** 5 sec **c** No

2

a $t = \frac{1}{4}, t = 2\frac{3}{4}$ **b** $2\frac{1}{2}$ sec **c** $t = 1\frac{1}{4}, t = 1\frac{3}{4}$ **d** $\frac{1}{2}$ sec

3

a $w = 10$ **b** 16 yards

4 $t = 4$

5

a $x = 10$ **b** Square of the number is 100.
17 times the number is 170.
$170 - 100 = 70$

6

a 6 in **b** 6.6 in

7

a $4x^2 + 44x$ m^2 **b** $x = 2$

8

a $m = 10$ **b** 15 hours

9

a $w = 2.33$ **b** 15.3 ft

10

a 120

b The amount of time it takes for the frisbee to reach the ground.

c 160 meters

11 3

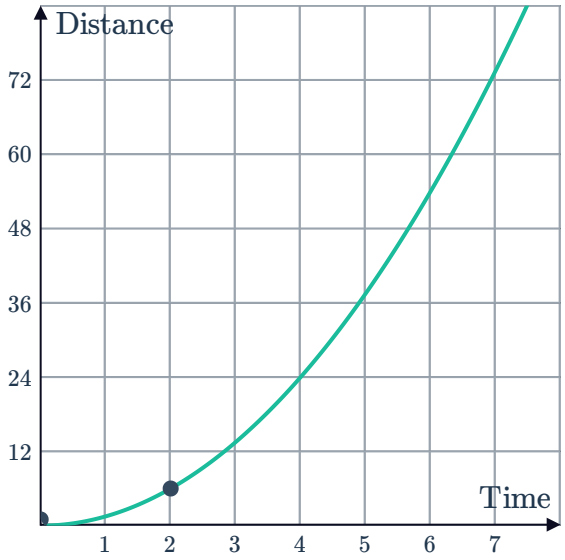
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12

a

Time	0	2	4	6
Distance	0	6	24	54

b



c $t = 2$

13

a 313.6 yards

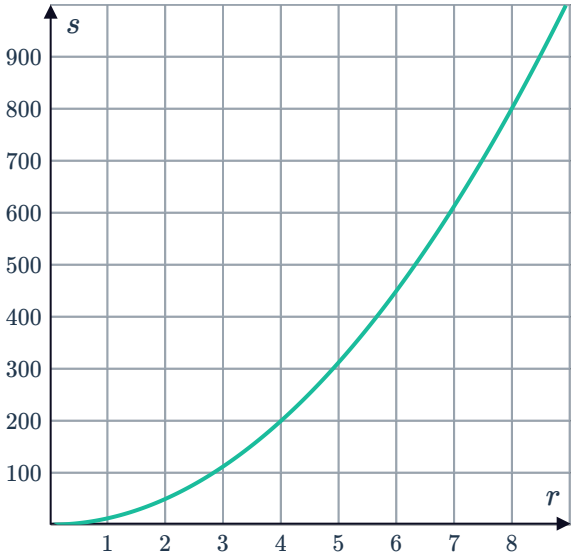
b $t = 4$

14

a

r	1	2	3	4	5	6	7	8
S	12.57	50.27	113.1	201.06	314.16	452.39	615.75	804.25

b



c

i $S = 380$

ii $r = 4$

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15

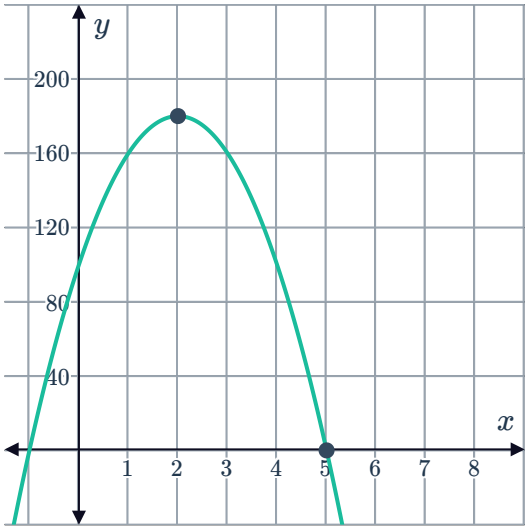
a

x	0	1	2	3	4	5	6
y	100	160	180	160	100	0	-140

b 5 seconds

c 180 feet

d



16

a $y = -\frac{1}{20}(x-18)^2 + 16.2$

b $4071.50\ \text{yd}^2$

17

a $A(0, 3.5), B(3, 0), C(7, 0)$

b 3

c $9a + k = \frac{7}{2}$

d $0 = 16a + k$

e $a = -\frac{1}{2}$

f 8 m

18

a $A(18, 0), B(0, 108)$

b $y = -\frac{1}{3}(x-18)(x+18)$

c $x = 9\sqrt{2}$

19

a $(180, 98)$

b $y = -\frac{49}{16\,200}(x-180)^2 + 98$

c Below

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20

a

t	0	1	2	3	5	6
d	0	16	64	144	400	576

b Initially, the object has not traveled any distance.

c Because time cannot be negative.

21

a $t = 3$ **b** $t = 9$

22 $t = 6$

23

a $m = 1900$ **b** $64.8\ \text{km/h}$ **c** $v = 16$

24

a $S = 528$ **b** $n = 5$

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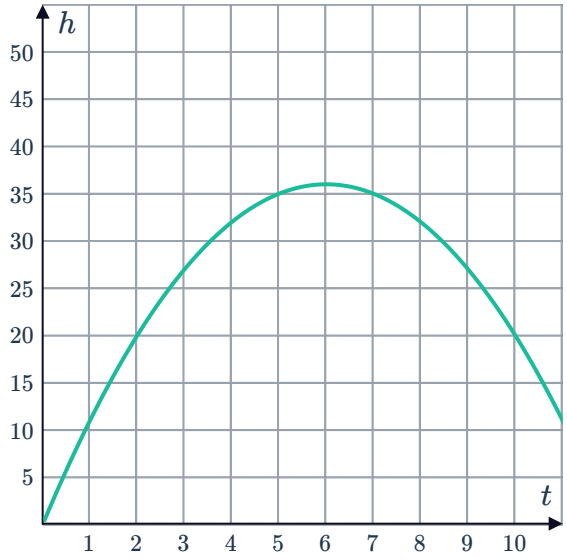
Maximum and minimum problems

25

a

t	1	2	3	4	5	6	7	8	9	10
h	11	20	27	32	35	36	35	32	27	20

b



c 33.75 meters

d 36 meters

26

a 58 **b** $80 - x$ **c** $y = x\,(80 - x)$ **d** 1600

e Yes

27

a 336 square meters **b** $32\ \text{m}$

c 400 square meters **d** Length $= 20\ \text{m}$, Width $= 20\ \text{m}$

28

a $y = x\,(24 - x)$ **b** $x = 12$ **c** 144

29

a $A = x\,(20 - x)$ **b** 10 units **c** 10 units

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30

a $s = 12$ **b** 337

31

a $t = 3$ **b** 122 units

32

a $t = 2.84$ s **b** $t = 1.38$ s **c** 34.25 m

33

a $13 - x$ **b** $x\,(13 - x)$ **c** 6.5 yd **d** 42.25 yd^2

34

a $(168 - x)$ m **b** $x\,(168 - x)$ m^2 **c** 84 m **d** 7056 m^2

35

a 16 **b** 64 meters **c** 16 meters **d** 39 meters

e $0 \leq d \leq 32$

36

a 576 feet **b** 6 seconds **c** 12 seconds

37

a 50 meters **b** $x = -10, 10$

c 37.50 meters **d** $x < -20$ and $x > 20$

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